Exact joins for the eight sporadic rows in the Higman–Sims table

Source recovery component

The fixed source is D. Leemans, *The residually weakly primitive geometries of the Higman–Sims group*, Australasian J. Combin. 33 (2005), 231–236, Table 1 on p. 232. We join existing complete computations and explicit source coordinates. No subgroup or geometry census is repeated. The last row retains the qualifications below; this component does not claim completion of the whole article.

1 Subgroup counts from complete retained inputs

For a representative H_i of a G-conjugacy class, its class length is $\ell_i = [G : N_G(H_i)]$. Thus the total number of subgroups is $\sum_i \ell_i$, provided that the representative list is complete. The six atlas inputs for $M_{11}, M_{12}, J_1, M_{22}, J_2, M_{23}$ have separately accepted complete numerical tables, with literal class-length and source-row correspondences. Their exact review records, all rows and required dependencies are acquired. No abstract name is inferred from an order or from matching the source total.

Proposition 1. *In a complete table of marks, let $m_{i,1}$ be the mark at the trivial subgroup and $m_{i,i}$ the diagonal mark. Then*

$$\ell_i = \frac{m_{i,1}}{m_{i,i}}.$$

Proof. The transitive coset set associated with H_i has $[G : H_i]$ elements, all fixed by the identity. Its H_i-fixed cosets are precisely those represented by elements of $N_G(H_i)$: a containment between conjugates of the same finite order is equality. Hence the two marks are $[G : H_i]$ and $[N_G(H_i) : H_i]$. Their quotient is $[G : N_G(H_i)]$. Distinct conjugacy classes partition the subgroup set. □

For HS the retained complete 589-class input is already used by its accepted geometry census. A bounded extraction checked equality of all 589 subgroup orders and the literal ordered generating permutations against that exact table model. It returned every pair of marks, and compared their ratios with the library class lengths. The clean terminal execution is retained; the present Python check independently recomputes every ratio and the sum. Thus this is a new arithmetic join to an acquired complete input, not a new subgroup enumeration.

G	$\lvert G\rvert$	conjugacy classes	subgroups
M_{11}	7920	39	8651
M_{12}	95040	147	214871
J_1	175560	40	158485
M_{22}	443520	156	941627
J_2	604800	146	1104344
M_{23}	10200960	204	17318406
HS	44352000	589	149985646

These 21 fields agree with the fixed Table 1. In particular the printed J_1 total is 158485; an earlier project transcription of that number is not a source correction.

2 Geometry populations and truncations

The retained panel-recognition and maximal-deletion induction enumerate firm, residually connected, flag-transitive, RWPRI coset geometries with $(\mathrm{IP})_2$. The local $(2T)_1$ condition is imposed on the actual panel actions. For the phase-six M_{11}, M_{12}, J_1 calculations it is applied to the complete broader census afterward; for the phase-seven M_{22}, M_{23} calculations it is an input filter. Every Borel anchor has either no admissible panels or a terminal empty level. A finite rank cutoff without such evidence is not used as exhaustion.

Equivalence in these five retained computations is under $\mathrm{Aut}(G)$ and arbitrary type permutation. Their actual whole-family transporters and finite recognition tests are retained. The M_{11} certificate is expressly acquired here with its full completed execution and source artifacts; the prior L096 assembly's five-item census inventory did not itself list this certificate. We do not invent a separate earlier acceptance of it. The common recognized method, subgroup catalogues and exact finite-group software are explicit premises of the present reuse.

A truncation removes types from a higher-rank member of the *same* locally two-transitive class. This restriction matters: allowing higher members of the broader non-$(2T)_1$ class gives different answers, for example the M_{11} rank-four value becomes 5 instead of 11. Such a broader operation is not silently substituted for Table 1's class. Exact retained proper-subset tests and positive family maps determine the deleted classes. The checker joins the full witness counts, $(2T)_1$ flags, fusion partitions and retained truncation outputs; it acquires the underlying complete tests rather than rerunning them.

G	rank 2	rank 3	rank 4	rank 5	rank 6
M_{11}	0	9	11	0	0
M_{12}	5	10	17	1	0
J_1	11	1	2	0	0
M_{22}	1	20	12	9	0
J_2	5	16	3	0	0
M_{23}	0	3	16	10	0
HS	4	20	41	8	1

These are nontruncation counts in the acquired equivalence conventions. The accepted J_2 join uses 87 proper subsets, ten exact positive maps and six truncation targets. Its complete population is $8, 19, 3$ in ranks $2, 3, 4$. The HS result uses its separately accepted bare-incidence classification, all 756 proper subsets, 15 positive maps and twelve distinct truncation targets; its complete population is $6, 30, 41, 8, 1$. The HS bare-separation argument is not transferred to other groups without proof.

The source prints 11 for M_{12} rank three, whereas the exact retained specified-action calculation gives 10 under the class-restricted truncation convention. We record this precise comparison; no assertion that a different, unproved bare-incidence quotient has ten classes is needed. The missing M_{12} archive class identified in the prior accepted work fails local two-transitivity, so it cannot repair this entry. No nontruncation count is obtained merely by subtracting the number of witness maps: distinct target classes, not map multiplicities, are removed.

3 The remaining Janko boundary

The separately retained J3 supplement is dated 28 November 2003 and has thirteen pages. All its native pages were read for this source join. Its 137 printed subgroup rows sum to 71564248, which agrees with Table 1. This is exact arithmetic on a transcribed table, not a new proof that the

subgroup roster is complete or that every printed normalizer is correct. The cited classification is attributed to Leemans's J3 article; the subgroup table cites the earlier subgroup work. The main article's full native PDF was not acquired during this audit.

The supplement explicitly says "up to isomorphism". It prints seven rank-two diagrams, of which two are identified as truncations, and seventeen rank-three diagram labels. Four rank-three labels carry a multiplicity two, giving 21 supplied occurrences. The accepted literal archive analysis instead has 19 classes under $\mathrm{Aut}(J_3)$ and type permutation, with the precise source-label correspondence retained. Labels 3.7 and 3.8 each retain two such classes; the pairs at 3.11 and 3.13 are fused by actual outer maps. Equality of diagrams does not itself identify geometries.

Accordingly we do not import "exactly 17 classes" as a completeness premise consistent with that literal population. Nor does the existence of 19 specified-action classes automatically prove 19 bare-incidence classes. A proposed completeness premise for the full *occurrence* list, and the full-group normalization needed to settle that equivalence boundary, require an explicit separate disposition. The rank-two value 5, the absence of higher-rank J3 geometries and the 137-class subgroup completeness remain source-attributed here. No fresh J3 census is claimed.

4 Executable and attribution boundary

The check rehashes all acquired inputs and recomputes the 732 accepted atlas row lengths, 589 HS mark ratios, five certificate joins and the accepted J_2/HS target-class complements. Fourteen damaged joins must fail. Literal mark-diagonal controls for C_2, C_2^2 and S_3 check the arithmetic interface. There is no GAP census in this replay. The old finite programs, retained catalogue assumptions, explicit reviews and source images remain bound. The inherited induction and recognition theory is reused, not advertised as a new theorem. No original-source, conjecture, Lean or compression credit follows from this component alone.